

Software in This Month's DVD

This month, we bring you a JTAG/boundary scan tool that is designed for newcomers. If you are a programming enthusiast, we have hardware description languages—PHDL and VisualHDL—the THDL++ based IDE tools for FPGAs

PANKAJ V.

goJTAG

With goJTAG software, you can study boundary scan techniques and carry out your own experiments, perform diagnosis and much more. It is a university-driven open source project which provides a full package for a JTAG newcomer including training materials, slides and exercises. The software also includes a simulation component that fully illustrates the underlying concept of JTAG/boundary scan and allows you to directly step-wise control the test features of IEEE 1149.1-compliant devices and observe a system's reaction in real time in an on-screen simulation. **Supported OS:** Windows, Unix, Linux **Link:** www.gojtag.com/sites/default/files/gojtag_v1.2j-Installer.exe

VisualHDL

VisualHDL is a new-generation integrated development environment (IDE) tool for rapid design of field-programmable gate arrays (FPGAs). It is based on THDL++, a C++ like language with fully parallel VHDL semantics. It allows you to develop fully parallel FPGA designs along with the flexibility, modularity and maintainability of C++ code, bringing the productivity of C++ and C++ IDEs to the FPGA world. **Supported OS:** Windows **Link:** <http://visualhdl.sysprogs.org/downloads/VisualHDL-2.1.exe>

SmartSim

SmartSim is a free and open source digital logic circuit design and simulation package. It is primarily intended for educational use by students and teachers, but can be used by anyone who wants to create and simulate digital logic circuits. It offers an interactive simulation feature which allows to control the circuit and explore inside the hierarchical

sub-components even when the circuit is running. **Supported OS:** Windows, Linux **Link:** http://smartsim.org.uk/download.php?item=win_x86_smartsim

IndigoSCADA

IndigoSCADA is an HMI design tool that allows you to use data for generating management reports. It is a powerful SCADA which is useful for small custom projects where MODBUS, DNP3, OPC DA 2.05, IEC 60870-5-101 and IEC 60870-5-104 need to be used. The HMI is based on QT library and you can insert and edit elements such as widgets, buttons and containers that can be customised and manipulated with ease. **Supported OS:** Windows **Link:** <http://sourceforge.net/projects/indigoscada/files/latest/download>

Scilab

Scilab is a free and open source software for numerical computation providing a powerful computing environment for engineering and scientific applications. This tool is used for learning engineering sciences and automatic control engineering. **Supported OS:** GNU/Linux, Mac OS, Windows **Link:** <http://www.scilab.org/download/5.4.1>

PHDL

PHDL is an open source research-based hardware description language (HDL) which models text-based schematics for printed circuit boards (PCBs). The language is compiled into a PCB netlist which can then be imported into a layout tool and can be used as an alternative to mainstream graphical schematic capture tools. PHDL aims at increasing designer productivity by overcoming the limitations of scale, collaboration and design reuse inherent with graphical schematic capture CAD tools. **Supported OS:** Windows, Linux, MacOS

Link: <http://sourceforge.net/projects/phdl/files/latest/download>

FreePCB

FreePCB is a free, open source Windows program for editing and laying out PCBs. It is an easy-to-learn and easy-to-use tool, yet having the capability of doing any professional quality work. It comprises a Footprint Wizard and Footprint Editor for creating or modifying footprints and have 1 to 16 copper layers. It outputs RS-274X-formatted Gerber file data and a drill file to make the boards of sizes up to 152.4cm×152.4cm. **Supported OS:** Windows **Link:** http://www.freepcb.com/downloads/freepcb_1200_setup.exe

Layout Editor

Layout Editor is a software to edit designs for MEMS and IC fabrication. You can also use it for multi-chip modules (MCMs), chip on board (COB), low temperature co-fired ceramics (LTCCs), monolithic microwave integrated circuits (MMICs), printed circuit boards (PCBs), thick film technology, thin film technology, amongst others. It supports a wide range of file formats. **Supported OS:** Windows, Linux, MacOS **Link:** <http://www.layouteditor.net/filedownload.php5?fileid=1432>

Kactus2

This DVD includes the latest release of Kactus2. It is the first graphical open source IP-XACT toolset to design embedded products. It has features for packaging IP for reuse and exchange, and can create 'electronic datasheets' of your existing IPs to simplify integration. It can also import, export and check for integrity of IP libraries from any standard compatible IP vendor. **Supported OS:** Windows, Linux **Link:** <http://sourceforge.net/projects/kactus2/files/Kactus2SetupWin64bit2.4.17.exe/download> ●
The author is a technical journalist at EFY